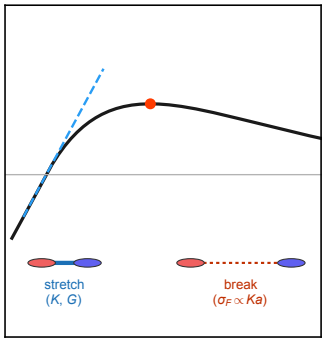
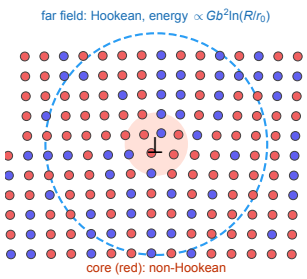
ATOMIC (DFT)

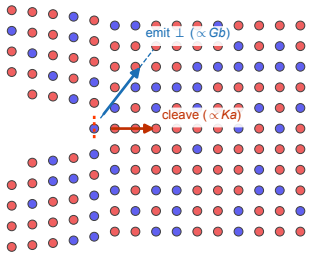
(a) bonds: Hookean vs breaking



(b) dislocation: core vs far field



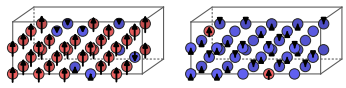
(c) crack tip: cleave or emit



α′ stiffer in shear (G ↑): emission harder, cleavage favoured

(d) DFT cells: α-like vs α′-like

40 atoms; arrows: QE local moments



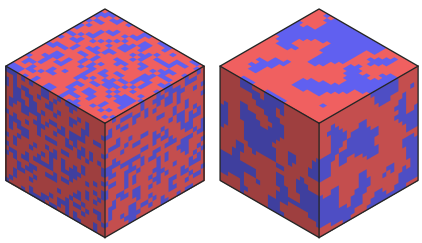
20% Cr SQS (QE) relaxed / exp. a: G = 96 / 88 GPa K/G = 1.98 / 1.81

95% Cr SQS (QE) relaxed / exp. a: G = 120 / 107 GPa K/G = 2.15 / 1.97

MOLECULAR (MD)

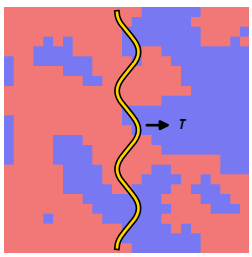
(e) MD cells: Fe−36Cr before and after decomposition

microstructure: 31,250 atoms (25^3 bcc), EAM



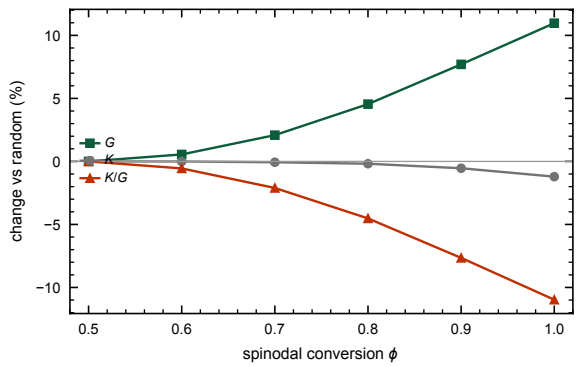
random (ϕ = 0.5) decomposed (ϕ = 1)

(f) α′ domains pin dislocations



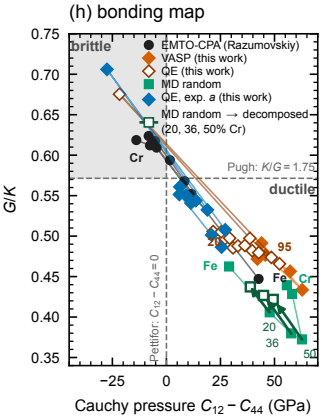
$G_{\alpha'} - G_{\alpha}$: +23 (EMTO-CPA), +15 (QE), +21 (QE exp. a), +5 GPa (MD); misfit 0.39%
⇒ modulus + misfit pinning: hardening

(g) MD: G rises, K flat, K/G falls with ϕ

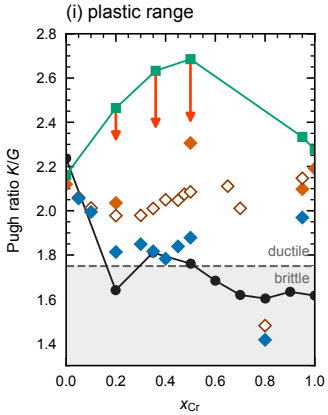


DATA & TGS

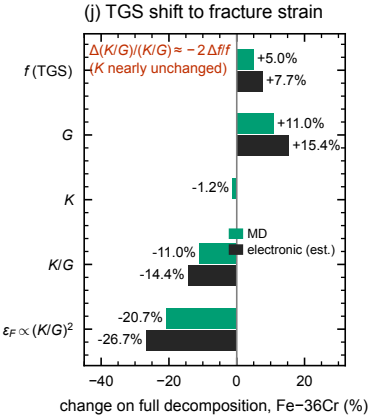
(h) bonding map



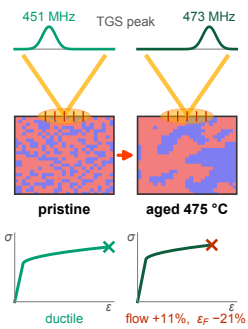
(i) plastic range



(j) TGS shift to fracture strain

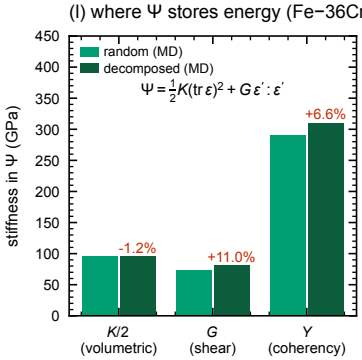


(k) what TGS sees

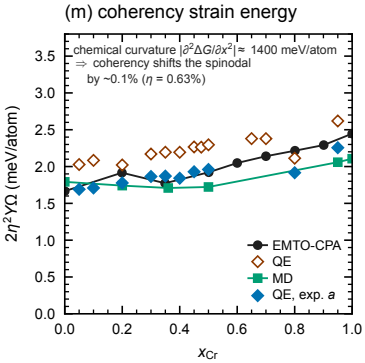


STRAIN ENERGY

(l) where Ψ stores energy (Fe−36Cr)



(m) coherency strain energy



(n) how it comes together

